



**CHALMERS**  
UNIVERSITY OF TECHNOLOGY

# MMS236 - Aircraft Design

Project 2025

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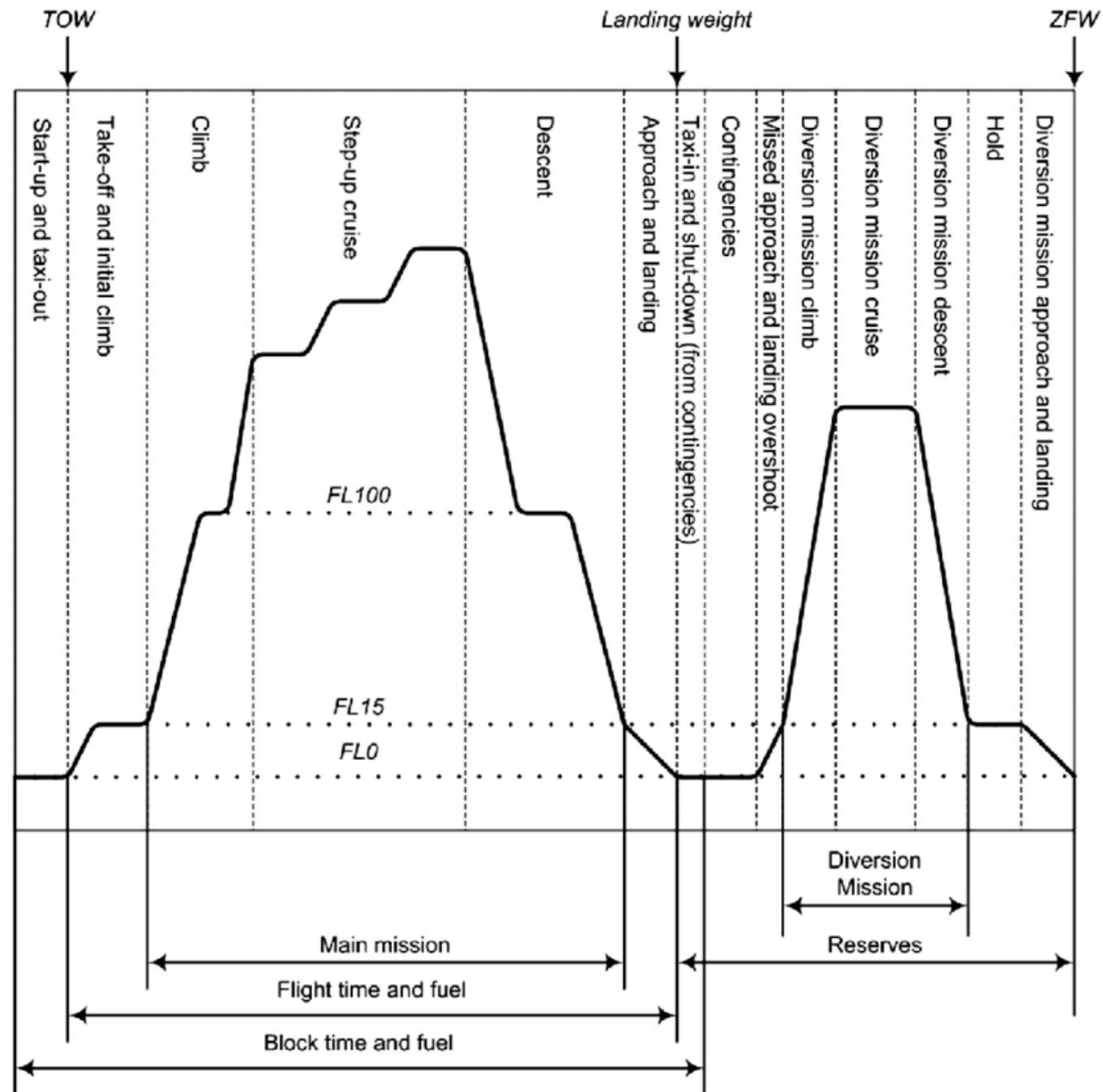
Carlos Xisto

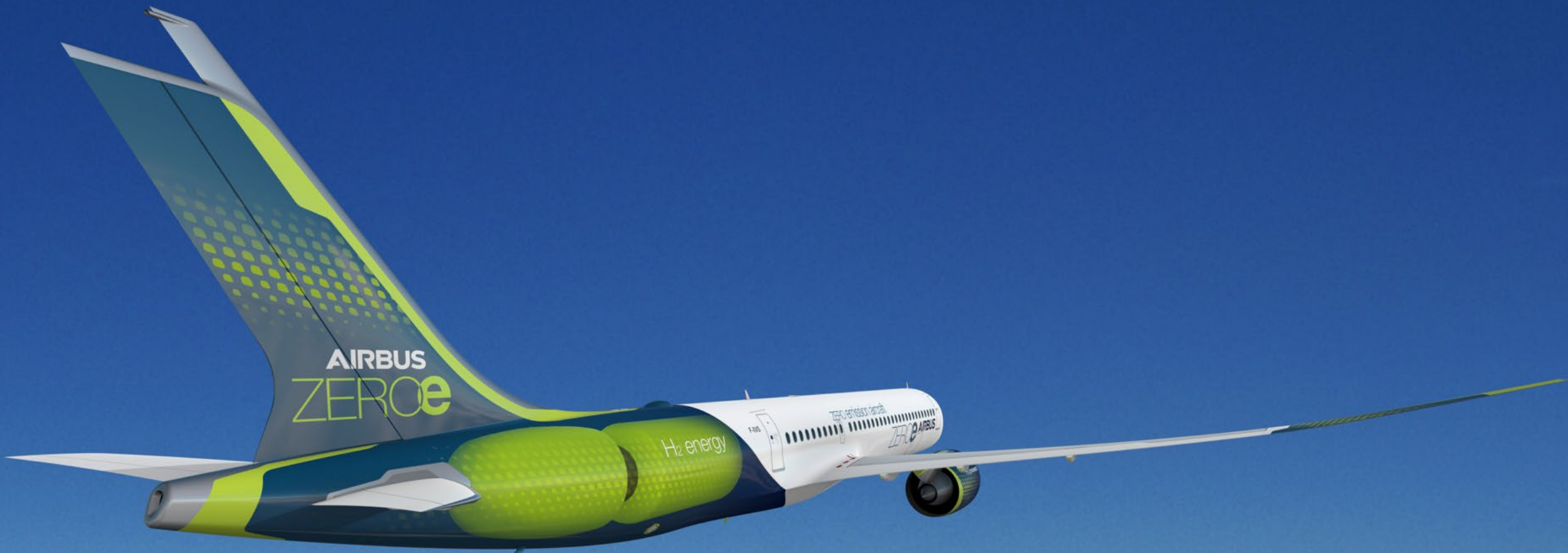
Division of Fluid Dynamics  
Department of Mechanics and Maritime Sciences

# CHALMERS



## Aero Spacelines Super Guppy





- Concept/purpose: Hydrogen powered Long range aircraft



EXAELIA

Flying Testbeds for Novel Long-Range Aircraft

|                               |                                |
|-------------------------------|--------------------------------|
| Fuel                          | Liquid hydrogen at 2 bar       |
| Range                         | 12500 km                       |
| Pax                           | 406 (economic) + 24 (Buisness) |
| Typical Passenger Weight      | 100 kg                         |
| Design Freight                | 10750 kg                       |
| Hold capacity                 | >206 m <sup>3</sup>            |
| Cruise speed                  | M0.85                          |
| Airports Compatibility Limits | F                              |
| Technology Freeze EIS-year    | 2050                           |



|                          |                                            |
|--------------------------|--------------------------------------------|
| TOFL (MTOW, SL, ISA)     | 2990 m                                     |
| Initial climb rate       | 2800 ft/min for a climb angle of 6 degrees |
| App Speed (MLW, SL, ISA) | 146 kts                                    |
| LFL (MLW, ISA)           | 2300 m                                     |



## ✈ Cruise

|                           |          |
|---------------------------|----------|
| Initial Cruise Altitude   | 35000 ft |
| Design Cruise Mach Number | 0.85     |
| Range                     | 12500 km |

## ✈ Reserves

|               |                          |
|---------------|--------------------------|
| Diversion     | 200 NM @ FL250 and M0.65 |
| Hold (loiter) | 30 min                   |
| Reserve fuel  | 3% contingency           |

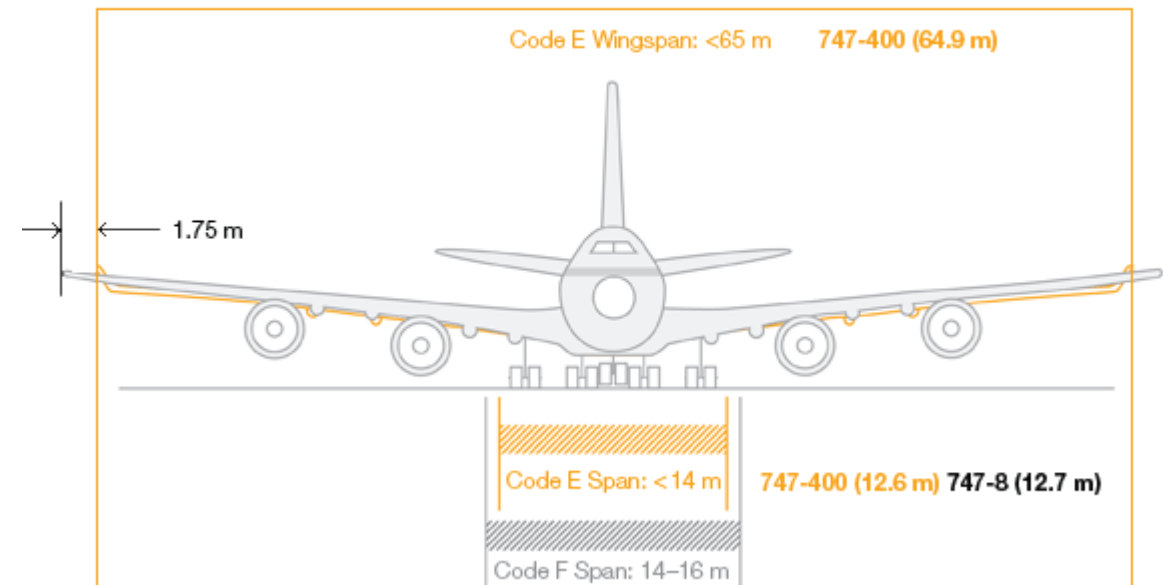
|        |               |
|--------|---------------|
| Code A | < 15 m        |
| Code B | 15 m - < 24m  |
| Code C | 24 m - < 36 m |
| Code D | 36 m - < 52 m |
| Code E | 52 m - < 65 m |
| Code F | 65 m - < 80 m |



Foldable tips can circumvent that, but folding and locking mechanism adds weight



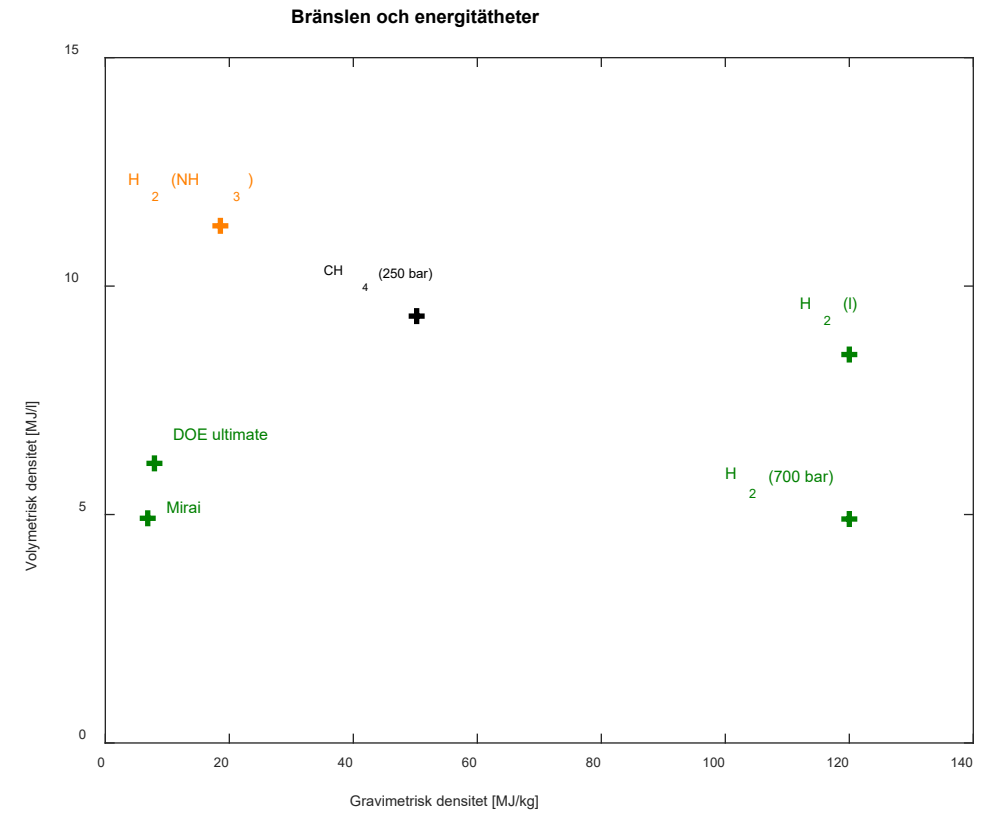
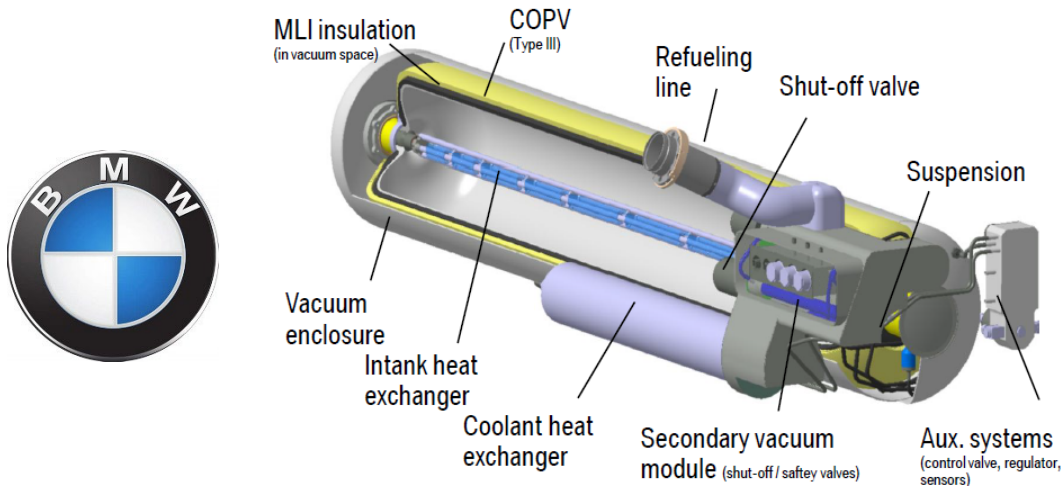
Code F Wingspan: 65–80 m 747-8 (68.4 m)



- Gas compressed

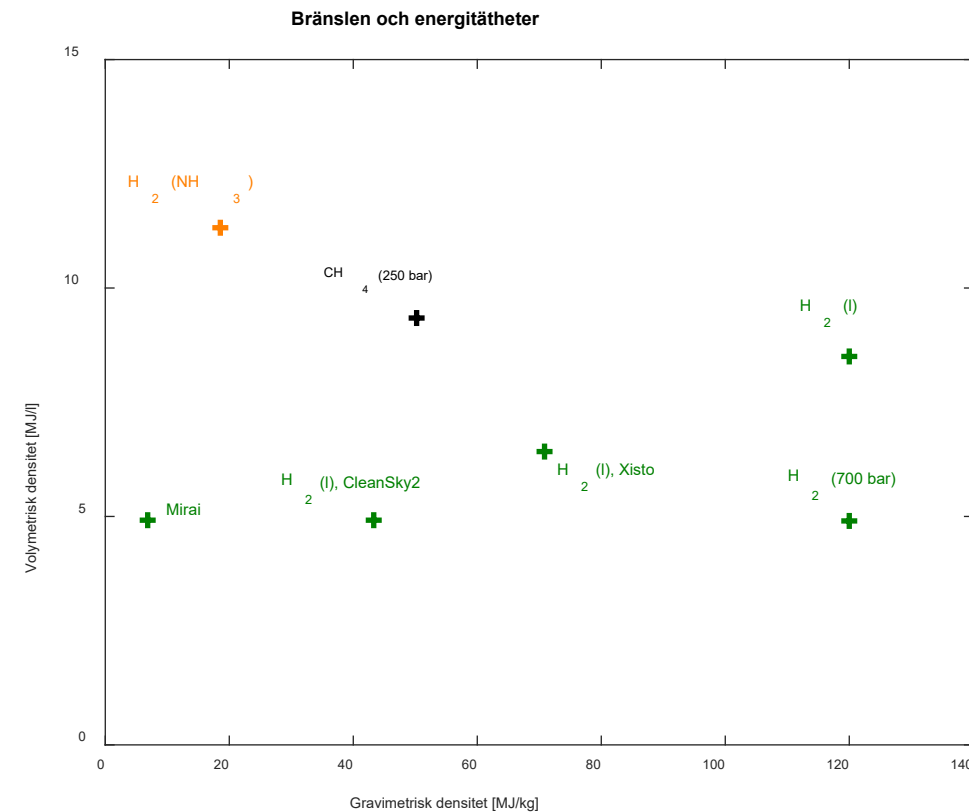
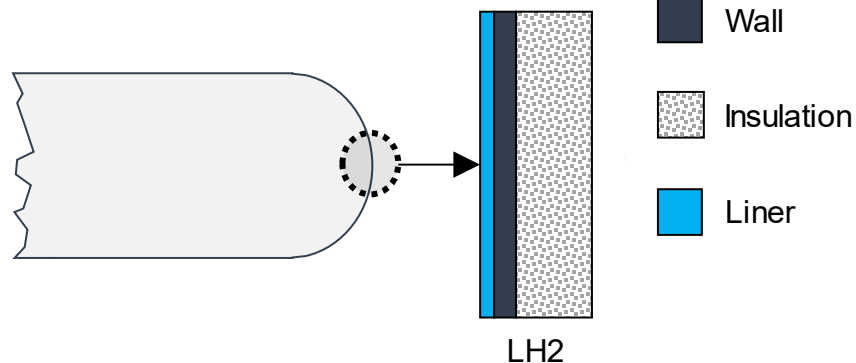


- Cryo-compressed



- ✈ Clean sky goal is a gravimetric density similar to jet-A
  - Fuel energy / (Tank mass + Fuel Mass)
- ✈ Insulated foam tanks are low TRL.
- ✈ Double wall vacuum tanks are heavy (difficult with longer ranges).
- ✈ Large uncertainty in gravimetric energy density.

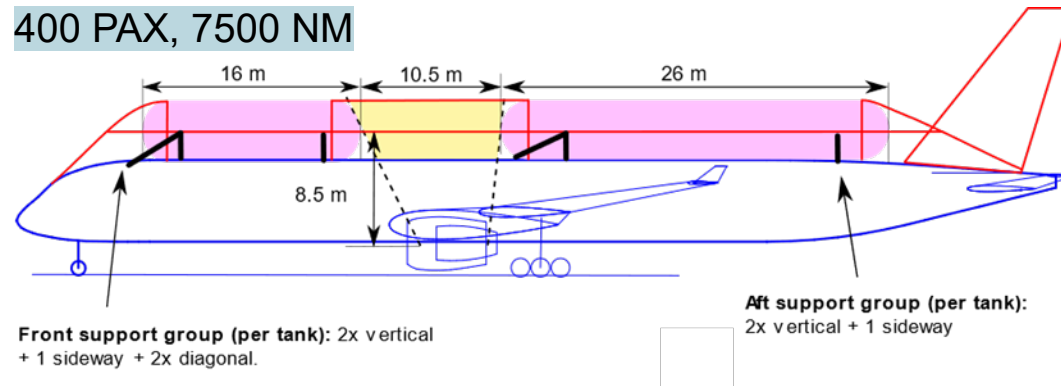
Polyvinylchloride  
rigid closed cell



$$G_i = \frac{\text{Fuel mass}}{\text{Fuel mass} + \text{tank mass}} = 0.5$$

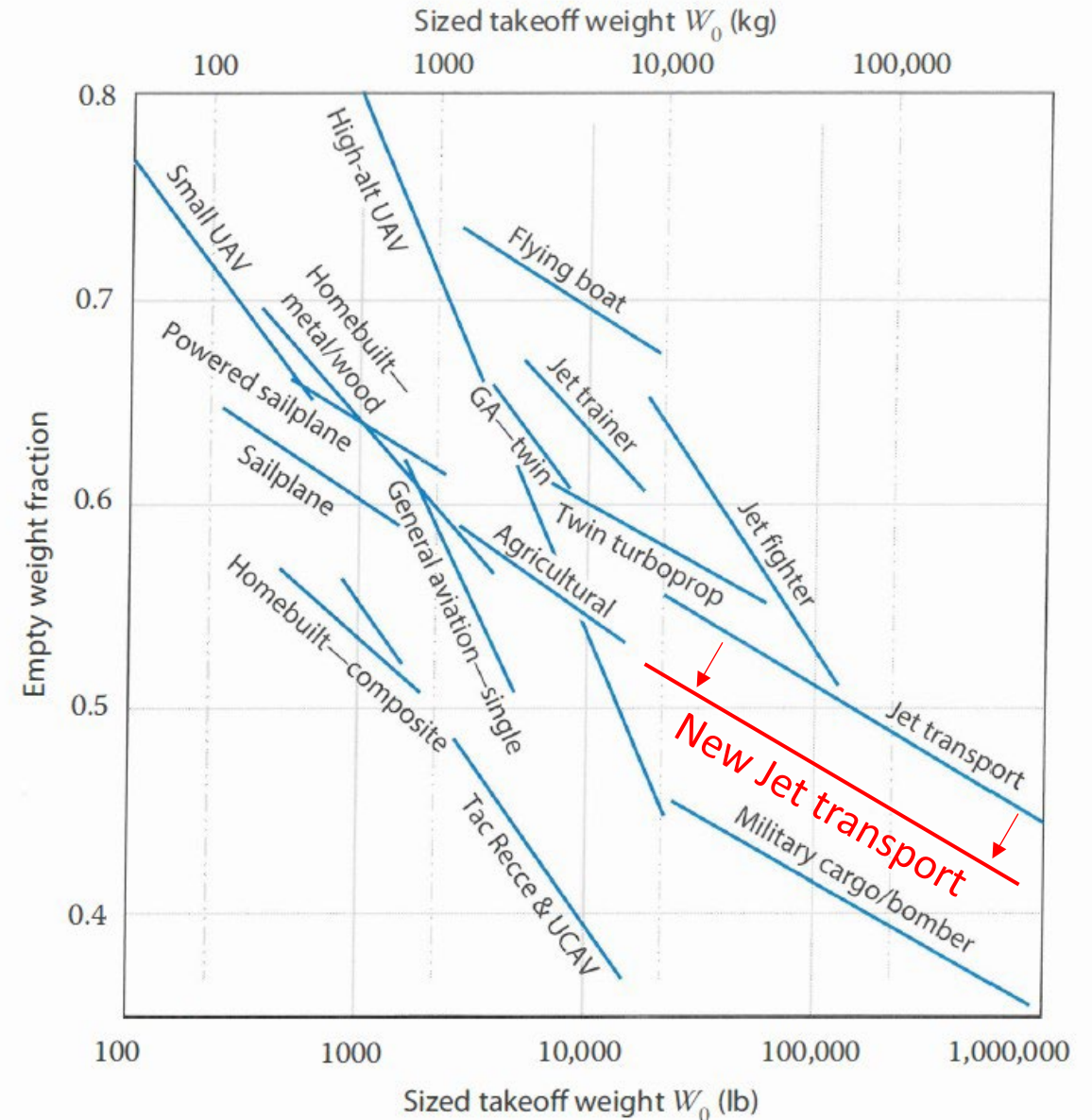
- ✈ ENABLEH2 project study.
- ✈ Chalmers concept follows simple existing idea.
- ✈ You are free to choose but be wise!

400 PAX, 7500 NM



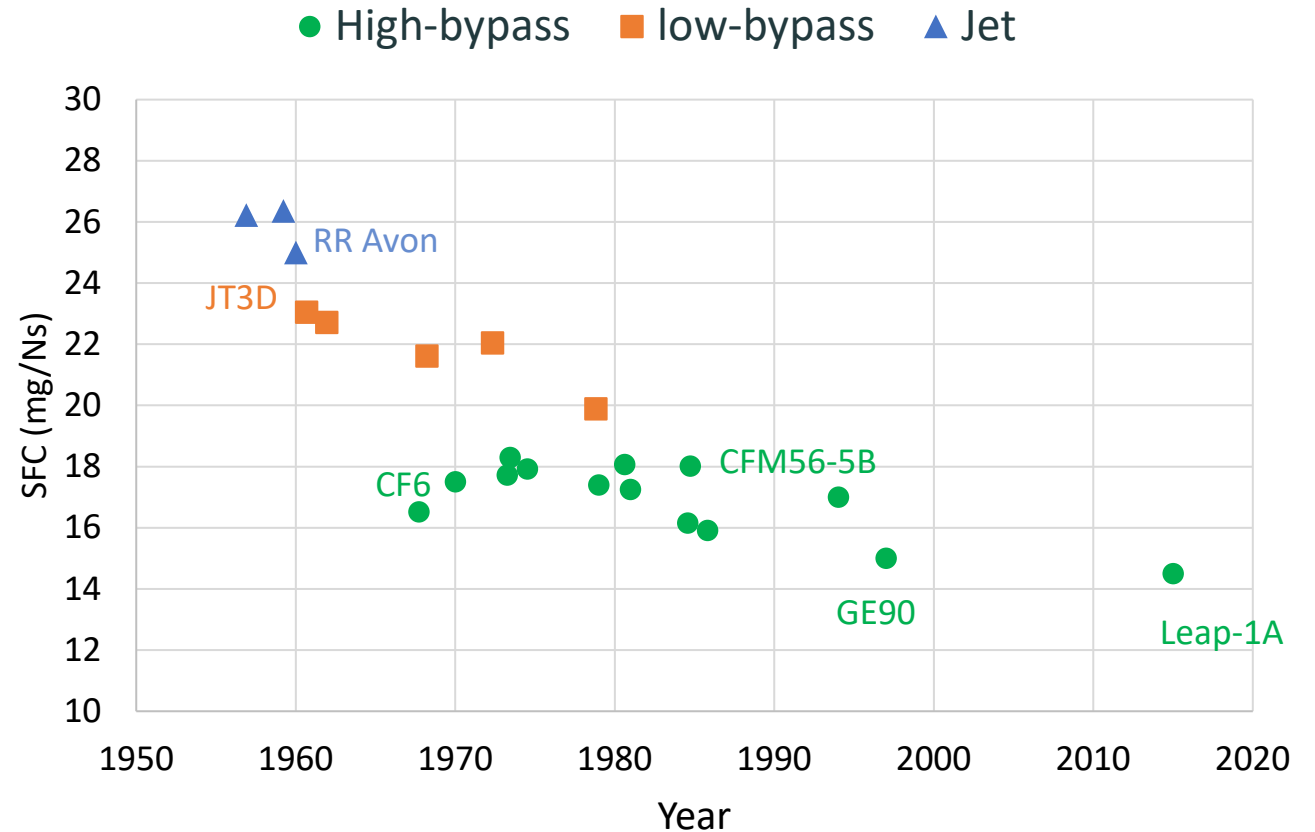
|                                                                             |                                                                              |                                                                          |                                                                                    |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <br>Baseline – low wings, tanks above fuselage                              | <br>1 – high wings, tanks above fuselage                                     | <br>2 – low wings, external tanks below wings                            | <br>3 – high wings, external tanks below wings                                     |
| <br>4 – low wings, external tanks above wings                               | <br>5 – external tanks joining box wings                                     | <br>6 – conformal tanks either side of fuselage and well below low wings | <br>7 – conformal tanks either side of fuselage and well below high wings          |
| <br>8 – low wings, tank inside aft fuselage                                 | <br>9 – high wings, tank inside aft fuselage                                 | <br>10 – low wings, tanks in forward and aft fuselage                    | <br>11 – high wings, tanks in forward and aft fuselage                             |
| <br>12 – low wings, tank in centre of fuselage                              | <br>13 – high wings, tank in centre of fuselage                              | <br>14 – low wings, tank aft of double-bubble fuselage                   | <br>15 – high wings, tank aft of double-bubble fuselage                            |
| <br>16 – low wings, tanks at forward and aft ends of double-bubble fuselage | <br>17 – high wings, tanks at forward and aft ends of double-bubble fuselage | <br>18 – low wings, tank(s) in centre of double-bubble fuselage          | <br>19 – high wings, tank(s) in centre of double-bubble fuselage                   |
| <br>20 – low wings, tanks aft and above double-deck fuselage                | <br>21 – mid-height wings, tanks aft and above double-deck fuselage          | <br>22 – low wings, tank(s) in centre of double-deck fuselage            | <br>23 – mid-height wings, tank(s) in centre of double-deck fuselage               |
| <br>24 – low wings, tanks at bottom of double-deck fuselage                 | <br>25 – mid-height wings, tanks at bottom of double-deck fuselage           | <br>26 – BWB with mid-height wings, internal under-floor tanks           | <br>27 – BWB with mid-height wings, int. under-floor and external under-wing tanks |
| <br>28 – low wings, double fuselages, tanks inside aft fuselages            | <br>29 – high wings, double fuselages, tanks inside aft fuselages            | <br>30 – high wings, double fuselages, external tank on centreline       |                                                                                    |

- ✈ Raymer correlations do not accurately represent modern aircraft
- ✈ Heavy insulated tanks for LH2 are not part of the correlations.
- ✈ What is a more accurate OEW/MTOW?
- ✈ What about L/D? The A320 might give  $L/D = 17$  at cruise. What about year 2050?
- ✈ Do some research.



- ✈ Typical values for cruise:
  - 25.5 mg/Ns for a turbojet
  - 22.7 mg/Ns for a low-bypass turbofan
  - 14.5 mg/Ns for high-bypass turbofan

- ✈ Make corrections for low-speed loiter:
  - 11.3 mg/Ns for high-bypass turbofan



Caution: Values are also impacted by the size and purpose of the engine (eg speed)

✈ Wetted aspect ratio:

$$A_{wet} = \frac{b^2}{S_{wet}} = \frac{A}{(S_{wet}/S_{ref})}$$

✈ Max  $L/D$  from statistics:

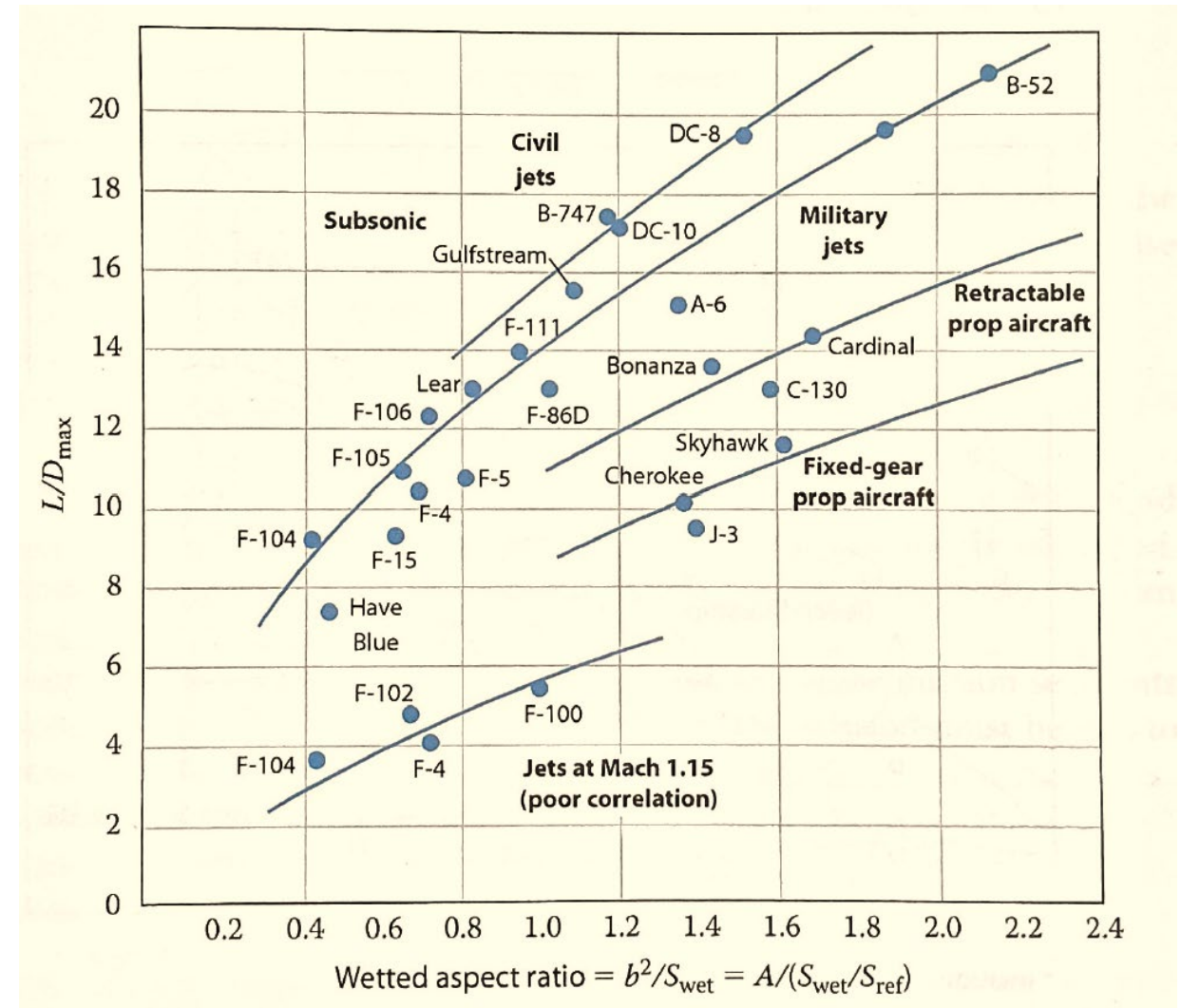
$$\left(\frac{L}{D}\right)_{max} = K_{LD} \sqrt{A_{wet}} = K_{LD} \sqrt{\frac{A}{(S_{wet}/S_{ref})}}$$

where:

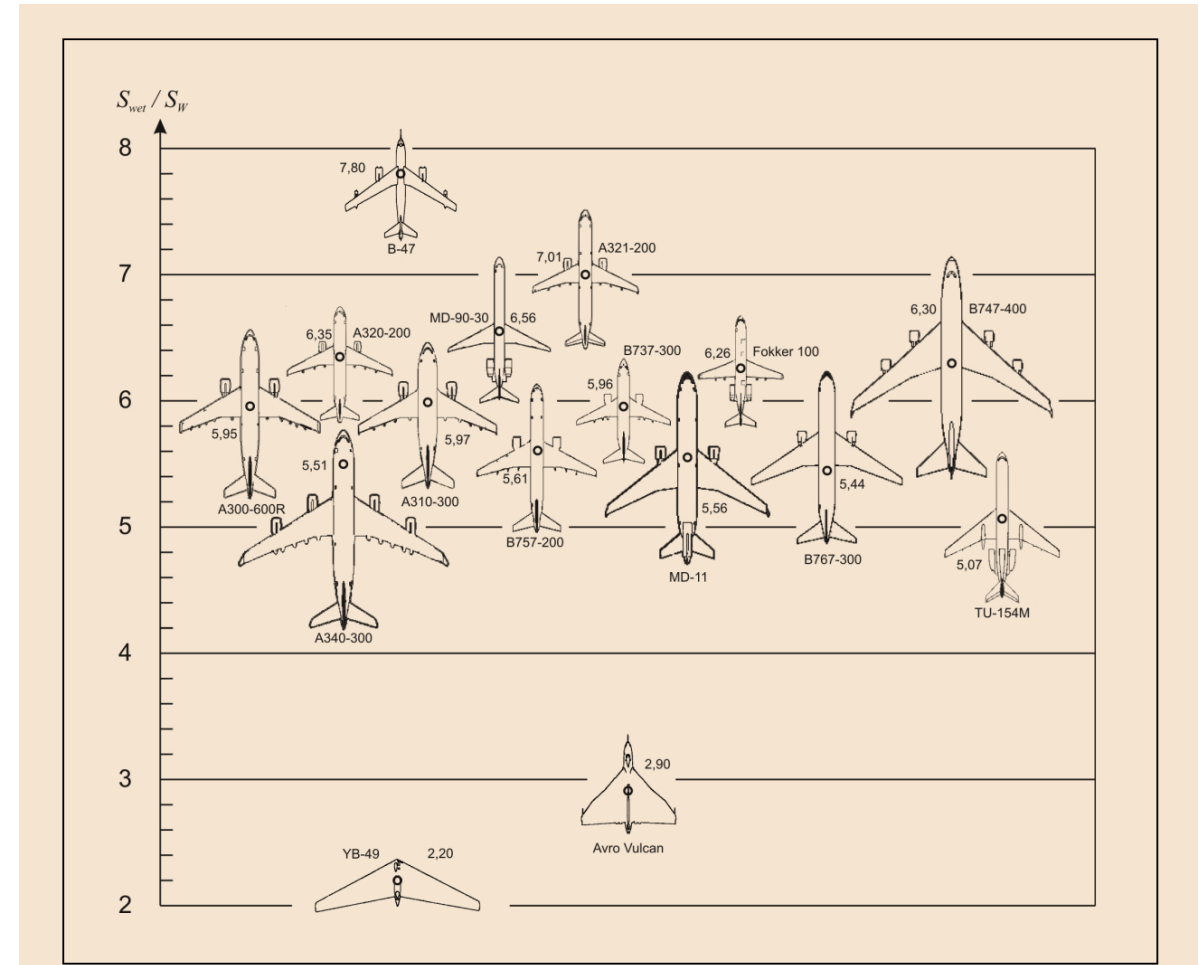
$$K_{LD} = 15.5 \text{ for civil jets}$$

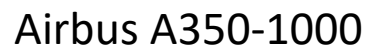
(coefs for other configurations available in the book)

What about  $S_{wet}/S_{ref}$ ? There is no design!



- ✈ For now we can "eyeball" by searching for similar designs.
- ✈ Drag varies with velocity and altitude.
- ✈ For maximum range:
  - Jet engine:  $0.866 \cdot \left(\frac{L}{D}\right)_{max}$
  - Piston - Propeller:  $\left(\frac{L}{D}\right)_{max}$
- ✈ For maximum loiter:
  - Jet engine:  $\left(\frac{L}{D}\right)_{max}$
  - Piston - Propeller:  $0.866 \cdot \left(\frac{L}{D}\right)_{max}$





# Thank You!



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